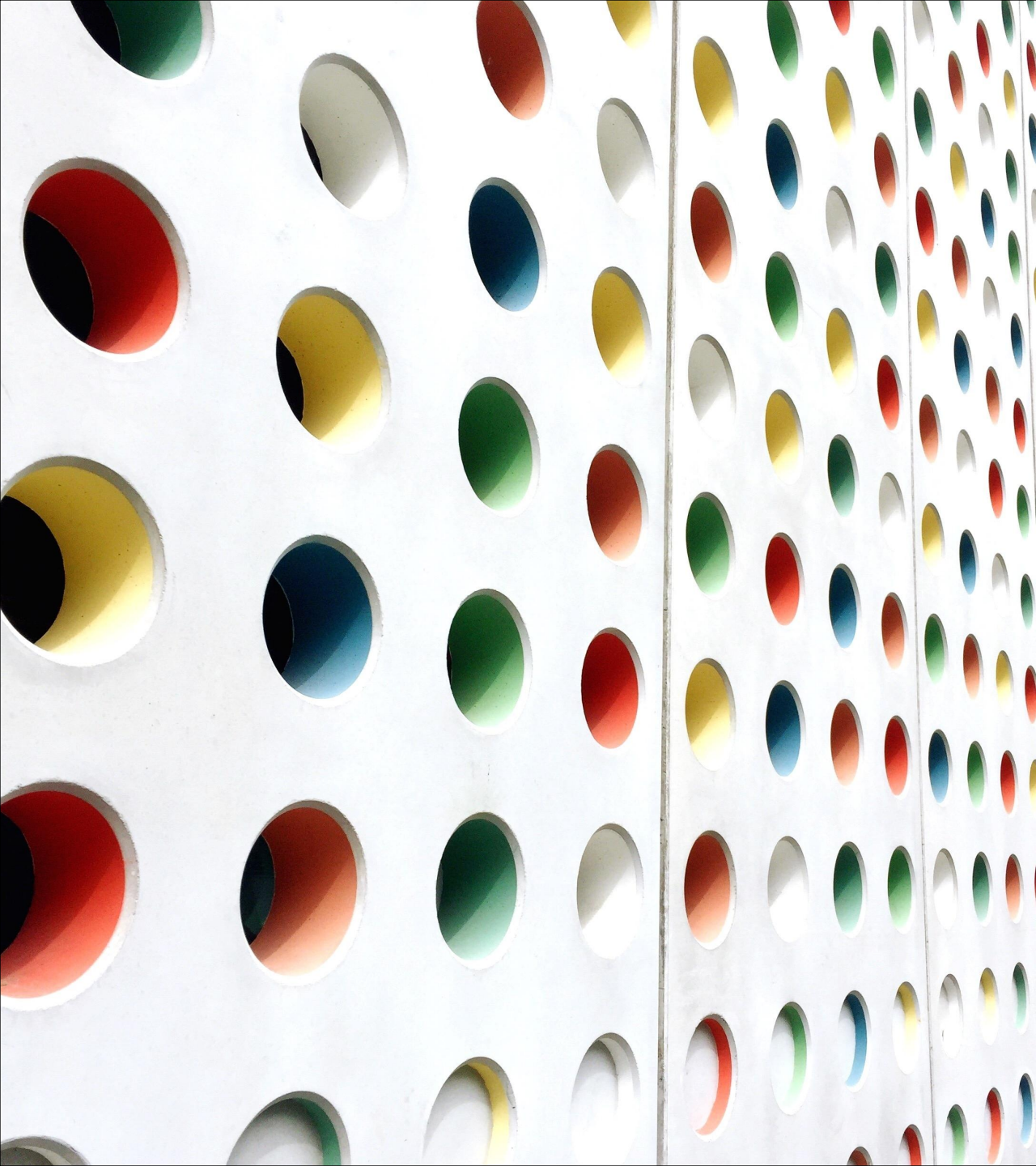
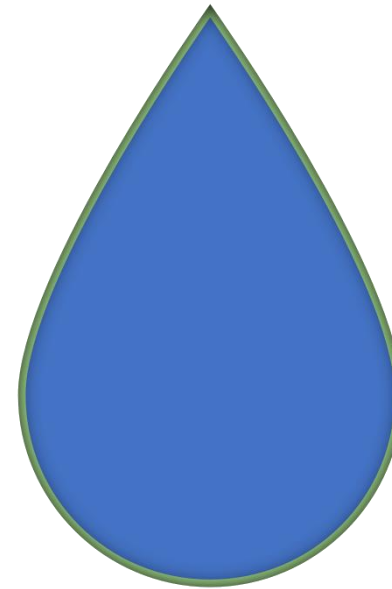




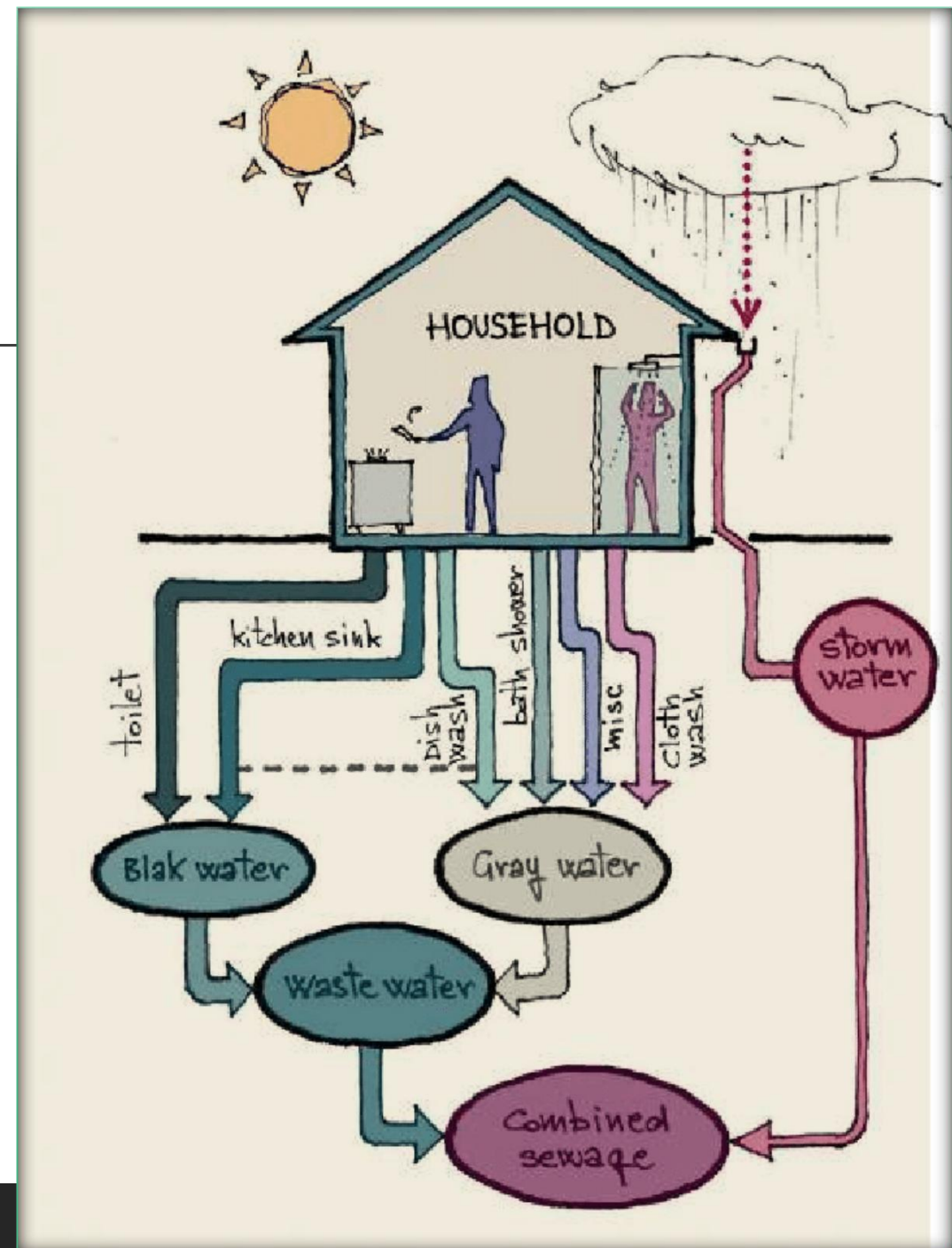
Wastewater quality and treatment

MODULE 3

Sources of wastewater from household



Sources of wastewater



Terminologies

Sullage/grey water - waste water from household sinks, showers, and baths, but not waste liquid or excreta from toilets.

Black water - contains human waste and is unsafe

Septage - excrement and other waste material contained in or removed from a septic tank

Night soil – human and animal excreta

Storm water

Sewage – domestic and industrial

Sewerage system

Characteristics of sewage

Physical characteristics of sewage

- Colour
 - Fresh sewage – light brown
 - Decomposing sewage – dark brown
 - Septic sewage – black
- Odour
 - Fresh sewage – no odour
 - Septic sewage – offensive odour
- Temperature
 - Slightly raised than water temperature
- Turbidity
 - Turbid depending on the quantity of solids
- Solids
 - 0.1% solids (suspended, dissolved or colloidal state)

Chemical characteristics of sewage

Fresh sewage is alkaline, putrefied sewage is acidic

- Carbonaceous and nitrogenous organic matter
- Fats and oils
- Persistent organic matters
- Inorganic matters

Chemical analysis

- pH
- Nitrogen
- DO
- BOD
- COD
- Chlorides
- Sulphates and sulphides
- Fats, Oil and grease
- Relative stability

Nitrogen

Indicates the presence of organic matter

Forms of nitrogen in wastewater

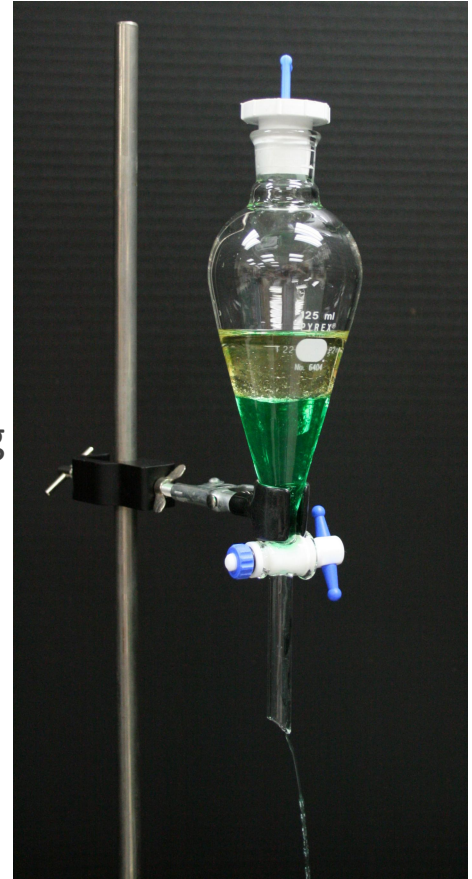
- Ammonia nitrogen or free ammonia – considerable amount of free ammonia indicates stale or old sewage
- Organic nitrogen – Kjeldahl method
- Albuminoid nitrogen – indicates the amount of undecomposed organic matter
- Nitrite nitrogen – indicates partly decomposed organic matter
- Nitrate nitrogen – indicates fully decomposed organic matter

Fats, oil and grease

From kitchen wastewater or industries like automobile workshops, factories etc.

Hinders biological oxidation process

- Measured by treating with dil. HCl and then dissolving in ether



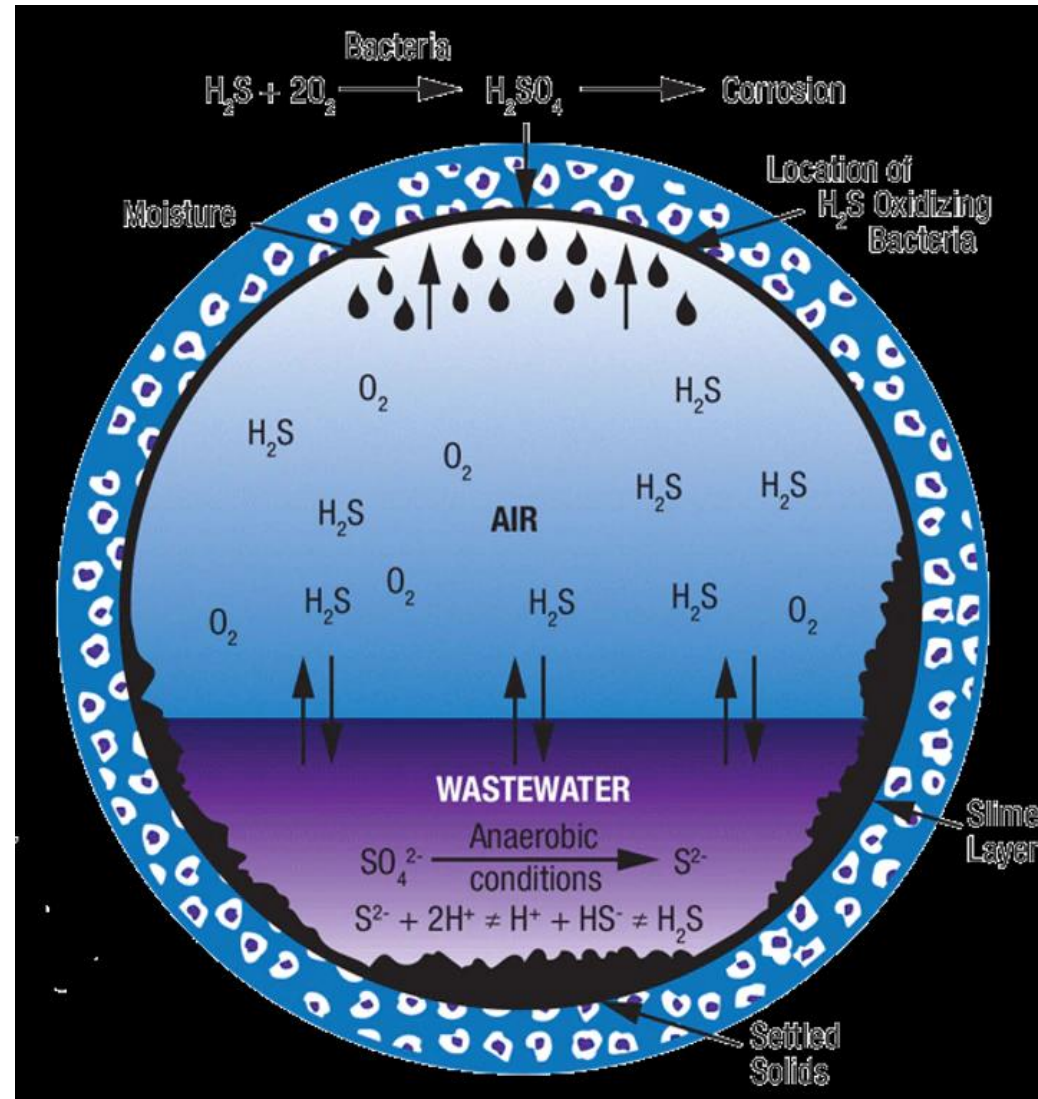
Sulphates and sulphides

Sulphates are naturally occurring

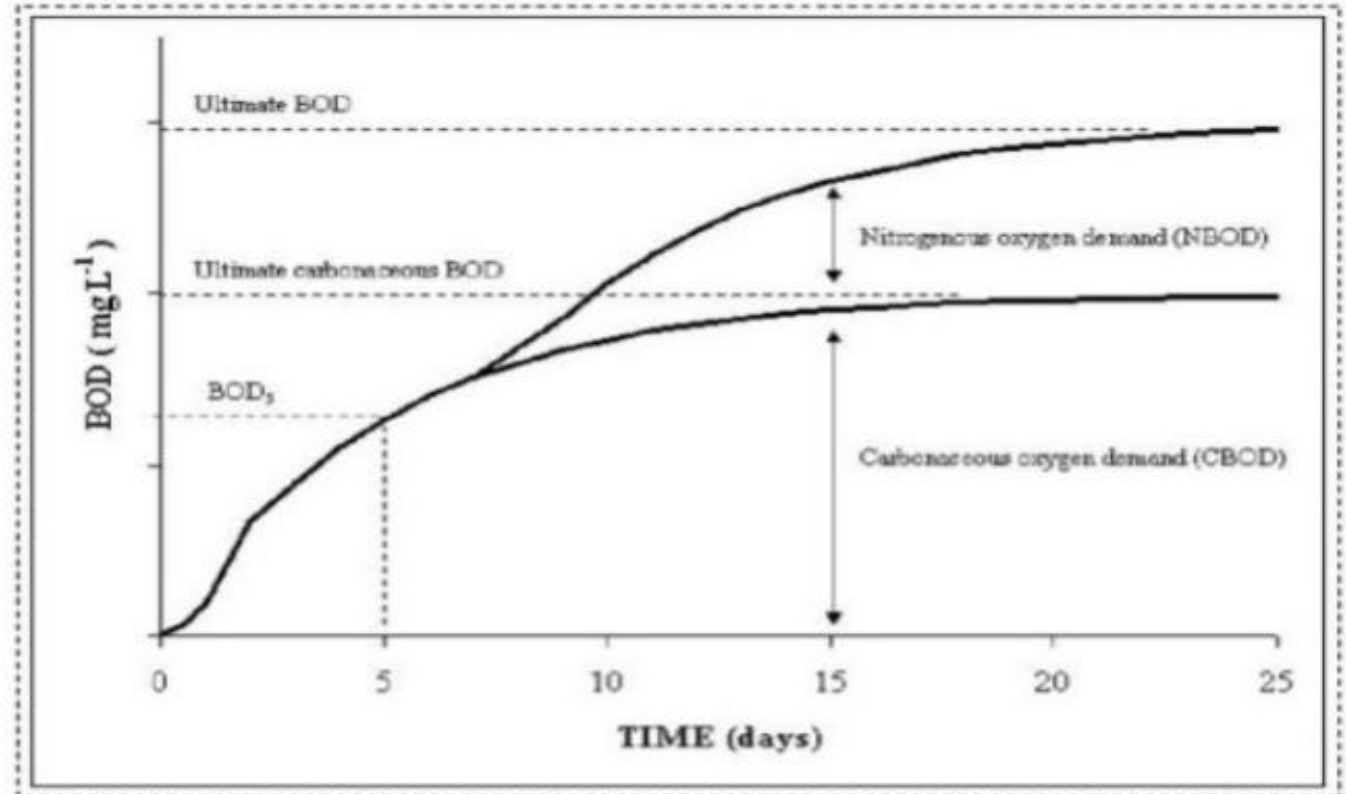
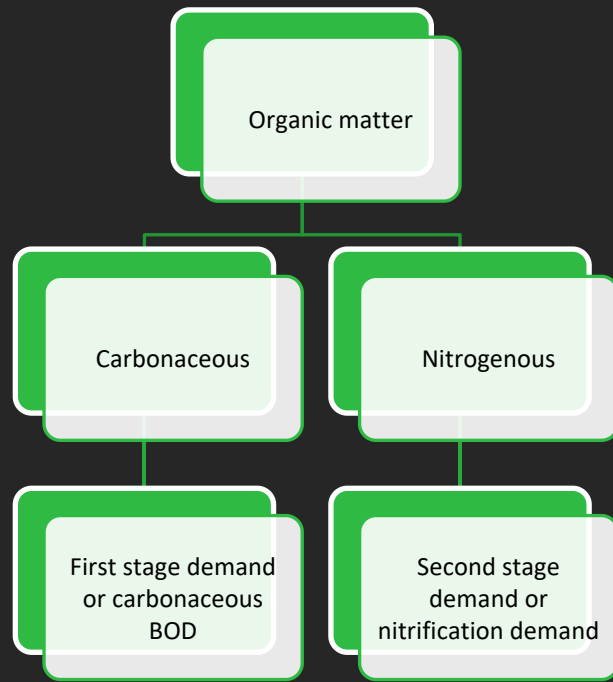
Converts to sulphides and H_2S by anaerobic decomposition

Sulphides in higher concentration affects the biological treatment processes

H_2S is corrosive to the sewer pipes (crown corrosion)



Oxygen demand



Oxygen demand

Ultimate Carbonaceous BOD – amount of oxygen necessary for the microorganisms in the sample to decompose the biodegradable carbonaceous materials

Total BOD = carbonaceous BOD + nitrogenous oxygen demand

Relative stability

Ratio of amount of oxygen available in the sewage to the amount of oxygen required to completely satisfy the first stage of BOD

Gives an idea about when the aerobic decomposition ends and anaerobic decomposition starts

Population equivalent

$$\text{population equivalent } P_e = \frac{\text{Total } BOD_5 \text{ of the industrial wastewater (Kg/day)}}{BOD_5 \text{ value per capita per day}}$$

Treatment of wastewater

- Physical methods – unit operations
 - Chemical methods
 - Biological methods
- } Unit Processes

Preliminary

Primary

Secondary

Tertiary

Typical layout for conventional sewage treatment plant

